

SYNERGY NETWORK

Remote Procedure Call (RPC) Specification

Canonical developer manual for the Synergy Network RPC surface

Part 2 of 2

Runtime, Governance, Cross-Chain and Operations

Document	Synergy Network Remote Procedure Call (RPC) Specification
Scope	Canonical developer manual for the Synergy Network RPC surface, with protocol-security-constrained semantics.
Coverage	Core protocol, identity and recovery, AIVM, governance, SXCP, analytics, operations, compatibility, and security-controlled interfaces.

Normative security baseline Methods whose legacy semantics would conflict with the Synergy security architecture are re-scoped here to the secure canonical form. Deployment policy is described through exposure profiles rather than environment nicknames.

Conventions inherited from Part 1

Part 2 uses the same JSON-RPC envelope, type system, authorization envelope, error model, and exposure profiles established in Part 1.

Unless a method is explicitly marked operator-profile or authority-plane only, state mutation in Part 2 is governed by the same authorization and simulation rules described in Part 1.

Part 2 coverage Advanced runtime, governance, cross-chain, analytics, operator, notification, and compatibility surfaces.

Node, Network and Subscription Interfaces

19 methods

Node health, peer topology, sync status, transport subscriptions, and operator-facing network control surfaces.

Methods: `synergy_nodeInfo`, `synergy_getNodeStatus`, `synergy_getSyncStatus`, `synergy_status`, `synergy_getNetworkStats`, `synergy_getPeerInfo`, `synergy_getNetworkVersion`, `synergy_getBootnodes`, `synergy_addPeer`, `synergy_removePeer`, `synergy_getPeerDetails`, `synergy_getNetworkDifficulty`, `synergy_getHashrate`, `synergy_getNodeDiscoveryTable`, `synergy_ping`, `synergy_getNodeUptime`, `synergy_subscribe`, `synergy_unsubscribe`, `synergy_subscription`

`synergy_nodeInfo`

```
synergy_nodeInfo()
```

Get node information.

Access profile: Public read profile.

Parameters

None.

Returns

- Node info object with:
 - name: Node name
 - version: Node version
 - protocolVersion: Protocol version (null if not set)
 - networkId: Network ID (338639 for network)
 - chainId: Chain ID (338639 for network)
 - consensus: Consensus algorithm
 - syncing: Sync status (boolean)
 - currentBlock: Current block number
 - timestamp: Current timestamp

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getNodeStatus`synergy_getNodeStatus()`

Get detailed node status including average block time and peer count.

Access profile: Public read profile.

Parameters

None.

Returns

- Status object with:
- node_type: Node type (null if not set)
- status: Node status ("running")
- uptime: Uptime percentage (string, e.g., "99.9%")
- uptime_seconds: Uptime in seconds
- version: Node version
- network: Network name
- sync_status: Sync status ("synced" or "syncing")
- last_block: Last block number
- avg_block_time: Average block time in seconds
- average_block_time: Alias for avg_block_time
- peers_connected: Number of connected peers
- peer_count: Alias for peers_connected
- peers: Alias for peers_connected
- timestamp: Current timestamp

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getSyncStatus`synergy_getSyncStatus()`

Get detailed synchronization status.

Access profile: Public read profile.

Parameters

None.

Returns

- Object with:
- syncing: Whether node is currently syncing
- current_block: Current block height
- highest_block: Highest known block in network
- starting_block: Block height when sync started
- sync_percentage: Sync progress percentage
- state: Current sync state ("Idle", "Discovering", "Downloading", "Validating", "Applying", "Synced")

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_status`synergy_status()`

Simple status check (legacy method).

Access profile: Public read profile.

Parameters

None.

Returns

"ok" string

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getNetworkStats`synergy_getNetworkStats()`

Get network statistics.

Access profile: Public read profile.

Parameters

None.

Returns

- Stats object with:
- `block_height`: Current block height
- `total_transactions`: Total number of transactions
- `active_validators`: Number of active validators
- `total_supply`: Total token supply
- `tokens`: Number of tokens
- `network_uptime`: Network uptime percentage
- `current_epoch`: Current epoch number
- `total_staked`: Total staked amount

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getPeerInfo`synergy_getPeerInfo()`

Get peer information.

Access profile: Public read profile.

Parameters

None.

Returns

- Object with:
- `peer_count`: Number of connected peers
- `peers`: Array of peer information objects with:
- `address`: Peer address

- `node_id`: Node ID
- `version`: Peer version
- `capabilities`: Peer capabilities
- `last_seen`: Last seen timestamp
- `blocks_sent`: Blocks sent
- `blocks_received`: Blocks received
- `txs_sent`: Transactions sent
- `txs_received`: Transactions received

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

`synergy_getNetworkVersion`

```
synergy_getNetworkVersion()
```

Get the network version/protocol version.

Access profile: Public read profile.

Parameters

None.

Returns

string - Network version

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

`synergy_getBootnodes`

```
synergy_getBootnodes()
```

Get the list of bootnodes.

Access profile: Public read profile.

Parameters

None.

Returns

- Array of bootnode objects with:
 - `address`: Bootnode address
 - `port`: Port number
 - `publicKey`: Public key
 - `status`: Online/offline status

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

`synergy_addPeer`

```
synergy_addPeer(enode)
```

Manually add a peer to connect to.

Access profile: Operator profile.

Parameters

- **enode** (*string*) — Enode URL of the peer

Returns

```
json
{
  "success": true,
  "message": "Peer added"
}
```

Security and execution rules

- Operator-profile method. Expose only on authenticated administrative control planes.
- If a deployment offers this method remotely, it SHOULD be protected by mutually authenticated transport, role-based policy, and audit logging.

synergy_removePeer

```
synergy_removePeer(peerId)
```

Remove a peer from the connection list.

Access profile: Operator profile.

Parameters

- **peerId** (*string*) — Peer ID

Returns

```
json
{
  "success": true,
  "message": "Peer removed"
}
```

Security and execution rules

- Operator-profile method. Expose only on authenticated administrative control planes.
- If a deployment offers this method remotely, it SHOULD be protected by mutually authenticated transport, role-based policy, and audit logging.

synergy_getPeerDetails

```
synergy_getPeerDetails(peerId)
```

Get detailed information about a specific peer.

Access profile: Public read profile.

Parameters

- **peerId** (*string*) — Peer ID

Returns

Detailed peer information including connection stats, capabilities, and history

Security and execution rules

- Read-only method. It does not change consensus state.

- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getNetworkDifficulty

```
synergy_getNetworkDifficulty()
```

Get the current network difficulty.

Access profile: Public read profile.

Parameters

None.

Returns

number – Current difficulty

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getHashrate

```
synergy_getHashrate()
```

Get the current network hashrate.

Access profile: Public read profile.

Parameters

None.

Returns

number – Current hashrate

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getNodeDiscoveryTable

```
synergy_getNodeDiscoveryTable()
```

Get the current Kademlia routing table.

Access profile: Public read profile.

Parameters

None.

Returns

Discovery table with peer buckets organized by distance

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_ping

```
synergy_ping()
```

Ping the node to check connectivity.

Access profile: Public read profile.

Parameters

None.

Returns

```
json
{
  "status": "ok",
  "latency_ms": 5
}
```

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getNodeUptime

```
synergy_getNodeUptime()
```

Get detailed node uptime statistics.

Access profile: Public read profile.

Parameters

None.

Returns

- Object with:
- totalUptimeSeconds: Total uptime in seconds
- uptimePercentage: Uptime percentage
- lastRestart: Last restart timestamp
- restartCount: Number of restarts
- averageSessionDuration: Average session duration

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_subscribe

```
synergy_subscribe(subscriptionType, newHeads, logs, pendingTransactions, ...)
```

Subscribe to events (WebSocket only).

Access profile: WebSocket subscription profile.

Parameters

- **subscriptionType** (*string*) — Type of subscription:
- newHeads: New block headers
- logs: Event logs
- pendingTransactions: Pending transactions
- validatorEvents: Validator events
- **filter** (*object, optional*) — Filter parameters

Returns

string – Subscription ID

Security and execution rules

- WebSocket-only streaming surface.
- Private or account-scoped streams SHOULD require the same authentication posture as the underlying method family.

synergy_unsubscribe

```
synergy_unsubscribe(subscriptionId)
```

Unsubscribe from events.

Access profile: WebSocket subscription profile.

Parameters

- **subscriptionId** (*string*) — Subscription ID

Returns

boolean - Success status

Security and execution rules

- WebSocket-only streaming surface.
- Private or account-scoped streams SHOULD require the same authentication posture as the underlying method family.

synergy_subscription

```
synergy_subscription()
```

Notification method for subscription events (server-to-client).

Access profile: WebSocket subscription profile.

Parameters

None.

Returns

Event data based on subscription type

Security and execution rules

- WebSocket-only streaming surface.
- Private or account-scoped streams SHOULD require the same authentication posture as the underlying method family.

AIVM Smart Contract Runtime

7 methods

Smart-contract deployment, execution, inspection, and source-verification methods for the AIVM runtime profile.

Methods: synergy_deployAIVMContract, synergy_executeAIVMContract, synergy_getAIVMContract, synergy_getAIVMContracts, synergy_getAIVMContractEvents, synergy_verifyAIVMContractSource, synergy_getAIVMStats

synergy_deployAIVMContract

```
synergy_deployAIVMContract(bytecode, abi, contractType, deployer, ...)
```

Deploy an AIVM smart contract.

Access profile: Authenticated client profile.

Parameters

- **bytecode** (*string*) — Contract bytecode (hex)
- **abi** (*string*) — Contract ABI (JSON string)
- **contractType** (*string*) — Type: "ai", "cross_chain", "oracle", "standard"
- **deployer** (*string*) — Deployer address
- **constructorArgs** (*array, optional*) — Constructor arguments

Returns

```
json
{
  "success": true,
  "contractAddress": "sync...",
  "transactionHash": "syntxn-...",
  "message": "Contract deployed successfully"
}
```

Security and execution rules

- State-mutating method. Successful execution is bound to the standard authorization envelope and consensus nonce tracking.
- The caller SHOULD obtain a preflight result from `synergy_simulateTransaction` or an equivalent domain-specific simulation step before submission.
- Off-chain signatures do not authorize state mutation on the Synergy Network RPC surface.

`synergy_executeAIVMContract`

```
synergy_executeAIVMContract(contractAddress, functionName, args, sender, ...)
```

Execute an AIVM contract function.

Access profile: Authenticated client profile.

Parameters

- **contractAddress** (*string*) — Contract address
- **functionName** (*string*) — Function to call
- **args** (*array*) — Function arguments
- **sender** (*string*) — Sender address
- **value** (*number, optional*) — Value to send
- **gasLimit** (*number, optional*) — Gas limit

Returns

Execution result with return values and gas used

Security and execution rules

- State-mutating method. Successful execution is bound to the standard authorization envelope and consensus nonce tracking.
- The caller SHOULD obtain a preflight result from `synergy_simulateTransaction` or an equivalent domain-specific simulation step before submission.
- Off-chain signatures do not authorize state mutation on the Synergy Network RPC surface.

synergy_getAIVMContract

```
synergy_getAIVMContract(address)
```

Get AIVM contract information.

Access profile: Public read profile.

Parameters

- **address** (*string*) — Contract address

Returns

- Contract object with:
- address: Contract address
- creator: Creator address
- bytecode: Contract bytecode
- abi: Contract ABI
- contractType: Type of contract
- deployedAt: Deployment timestamp
- state: Current contract state

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getAIVMContracts

```
synergy_getAIVMContracts(filter, limit)
```

Get all deployed AIVM contracts.

Access profile: Public read profile.

Parameters

- **filter** (*object, optional*) — Filter by type, creator, etc.
- **limit** (*number, optional*) — Maximum results

Returns

Array of contract objects

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getAIVMContractEvents

```
synergy_getAIVMContractEvents(contractAddress, eventName, fromBlock, toBlock)
```

Get events emitted by an AIVM contract.

Access profile: Public read profile.

Parameters

- **contractAddress** (*string*) — Contract address
- **eventName** (*string, optional*) — Specific event name
- **fromBlock** (*number, optional*) — Starting block
- **toBlock** (*number, optional*) — Ending block

Returns

Array of event objects

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_verifyAIVMContractSource

```
synergy_verifyAIVMContractSource(contractAddress, sourceCode, compilerVersion, optimization)
```

Verify and store AIVM contract source code.

Access profile: Authenticated client profile.

Parameters

- **contractAddress** (*string*)— Contract address
- **sourceCode** (*string*)— Contract source code
- **compilerVersion** (*string*)— Compiler version
- **optimization** (*boolean*)— Optimization enabled

Returns

```
json
{
  "success": true,
  "message": "Contract source verified"
}
```

Security and execution rules

- State-mutating method. Successful execution is bound to the standard authorization envelope and consensus nonce tracking.
- The caller SHOULD obtain a preflight result from `synergy_simulateTransaction` or an equivalent domain-specific simulation step before submission.
- Off-chain signatures do not authorize state mutation on the Synergy Network RPC surface.

synergy_getAIVMStats

```
synergy_getAIVMStats()
```

Get AIVM runtime statistics.

Access profile: Public read profile.

Parameters

None.

Returns

- Stats object with:
 - **totalContracts**: Total deployed contracts
 - **activeContracts**: Currently active contracts
 - **totalExecutions**: Total contract executions
 - **gasUsed**: Total gas used by contracts
 - **supportedFeatures**: List of supported features

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

Distributed AI Execution

9 methods

Distributed AI job submission, status, result, model discovery, validator task, and reward interfaces.

Methods: `synergy_initiateDistributedAI`, `synergy_getDistributedAIStatus`, `synergy_getDistributedAIResult`, `synergy_submitAIPartialResult`, `synergy_getValidatorAITasks`, `synergy_getValidatorAIRewards`, `synergy_getAIDistributedStats`, `synergy_getAIModels`, `synergy_chatWithAIVM`

`synergy_initiateDistributedAI`

```
synergy_initiateDistributedAI(modelId, inputData, clusterId, rewardPool)
```

Initiate a distributed AI computation.

Access profile: Authenticated client profile.

Parameters

- `modelId` (*string*) — AI model identifier
- `inputData` (*string*) — Input data (hex)
- `clusterId` (*number, optional*) — Specific cluster
- `rewardPool` (*number*) — Reward pool for computation

Returns

```
json
{
  "success": true,
  "computationId": "...",
  "assignedCluster": 1,
  "estimatedCompletionTime": 1640995200
}
```

Security and execution rules

- State-mutating method. Successful execution is bound to the standard authorization envelope and consensus nonce tracking.
- The caller SHOULD obtain a preflight result from `synergy_simulateTransaction` or an equivalent domain-specific simulation step before submission.
- Off-chain signatures do not authorize state mutation on the Synergy Network RPC surface.

`synergy_getDistributedAIStatus`

```
synergy_getDistributedAIStatus(computationId)
```

Get status of a distributed AI computation.

Access profile: Public read profile.

Parameters

- `computationId` (*string*) — Computation ID

Returns

- Status object with:
- `status`: Current status
- `progress`: Progress percentage
- `participatingValidators`: Array of participating validators

- `submittedPartialResults`: Number of partial results
- `estimatedCompletionTime`: Estimated completion time

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

`synergy_getDistributedAIResult`

```
synergy_getDistributedAIResult(computationId)
```

Get the result of a completed distributed AI computation.

Access profile: Public read profile.

Parameters

- `computationId` (*string*) — Computation ID

Returns

- Result object with:
- `result`: Computation result
- `aggregationMethod`: Method used to aggregate results
- `participatingValidators`: Array of validators that contributed
- `completionTime`: Actual completion time

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

`synergy_submitAIPartialResult`

```
synergy_submitAIPartialResult(computationId, validatorAddress, partialResult, proof)
```

Submit a partial result for a distributed AI computation.

Access profile: Authenticated client profile.

Parameters

- `computationId` (*string*) — Computation ID
- `validatorAddress` (*string*) — Validator address
- `partialResult` (*string*) — Partial result (hex)
- `proof` (*string*) — Proof of correct computation

Returns

```
json
{
  "success": true,
  "message": "Partial result submitted"
}
```

Security and execution rules

- State-mutating method. Successful execution is bound to the standard authorization envelope and consensus nonce tracking.
- The caller SHOULD obtain a preflight result from `synergy_simulateTransaction` or an equivalent domain-specific simulation step before submission.
- Off-chain signatures do not authorize state mutation on the Synergy Network RPC surface.

synergy_getValidatorAITasks

```
synergy_getValidatorAITasks(validatorAddress)
```

Get pending AI tasks for a validator.

Access profile: Public read profile.

Parameters

- **validatorAddress** (*string*) — Validator address

Returns

Array of task objects

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getValidatorAIRewards

```
synergy_getValidatorAIRewards(validatorAddress)
```

Get AI computation rewards for a validator.

Access profile: Public read profile.

Parameters

- **validatorAddress** (*string*) — Validator address

Returns

Rewards object with total and breakdown

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getAIDistributedStats

```
synergy_getAIDistributedStats()
```

Get distributed AI network statistics.

Access profile: Public read profile.

Parameters

None.

Returns

- Stats object with:
 - totalComputations: Total AI computations
 - completedComputations: Completed computations
 - activeComputations: Currently active
 - totalRewardsDistributed: Total AI rewards paid
 - averageCompletionTime: Average completion time
 - activeValidators: Validators participating in AI

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getAIModels

```
synergy_getAIModels(category)
```

Get available AI models for distributed computation.

Access profile: **Public read profile.**

Parameters

- **category** (*string, optional*) — Model category filter

Returns

- Array of model objects with:
- **modelId**: Model identifier
- **name**: Model name
- **description**: Model description
- **inputFormat**: Expected input format
- **outputFormat**: Output format
- **complexity**: Computational complexity
- **averageCompletionTime**: Average time to complete

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_chatWithAIVM

```
synergy_chatWithAIVM(contractAddress, message, context)
```

Interact with an AI-powered smart contract.

Access profile: **Authenticated client profile.**

Parameters

- **contractAddress** (*string*) — AI contract address
- **message** (*string*) — User message
- **context** (*object, optional*) — Conversation context

Returns

AI response object

Security and execution rules

- State-mutating method. Successful execution is bound to the standard authorization envelope and consensus nonce tracking.
- The caller SHOULD obtain a preflight result from `synergy_simulateTransaction` or an equivalent domain-specific simulation step before submission.
- Off-chain signatures do not authorize state mutation on the Synergy Network RPC surface.

Governance and Treasury

9 methods

Proposal, vote, execution, treasury, and voting-power surfaces. Governance counting remains subject to Synergy holding-period and snapshot rules.

Methods: `synergy_getProposals`, `synergy_getProposal`, `synergy_createProposal`, `synergy_voteProposal`, `synergy_executeProposal`, `synergy_getVotingPower`, `synergy_delegateVotingPower`, `synergy_getTreasuryBalance`, `synergy_proposeTreasurySpend`

`synergy_getProposals`

```
synergy_getProposals(status, limit)
```

Get all governance proposals.

Access profile: Public read profile.

Parameters

- **status** (*string, optional*) — Filter by status: "active", "passed", "rejected", "executed"
- **limit** (*number, optional*) — Maximum results

Returns

Array of proposal objects

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

`synergy_getProposal`

```
synergy_getProposal(proposalId)
```

Get a specific governance proposal.

Access profile: Public read profile.

Parameters

- **proposalId** (*string*) — Proposal ID

Returns

- Proposal object with:
- `proposalId`: Proposal ID
- `title`: Proposal title
- `description`: Proposal description
- `proposer`: Proposer address
- `status`: Current status
- `votesFor`: Votes in favor
- `votesAgainst`: Votes against
- `votesAbstain`: Abstain votes
- `votingStartTime`: Voting start timestamp
- `votingEndTime`: Voting end timestamp
- `executionTime`: Execution timestamp (if passed)

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

`synergy_createProposal`

```
synergy_createProposal(title, description, actions, proposer, ...)
```

Create a new governance proposal.

Access profile: Authenticated client profile.

Parameters

- **title** (*string*) — Proposal title
- **description** (*string*) — Proposal description
- **actions** (*array*) — Proposed actions
- **proposer** (*string*) — Proposer address
- **deposit** (*number*) — Proposal deposit amount

Returns

```
json
{
  "success": true,
  "proposalId": "...",
  "message": "Proposal created"
}
```

Security and execution rules

- State-mutating method. Successful execution is bound to the standard authorization envelope and consensus nonce tracking.
- The caller SHOULD obtain a preflight result from `synergy_simulateTransaction` or an equivalent domain-specific simulation step before submission.
- Off-chain signatures do not authorize state mutation on the Synergy Network RPC surface.

synergy_voteProposal

```
synergy_voteProposal(proposalId, voter, vote, signature)
```

Cast a governance vote under snapshot and holding-period rules enforced by the protocol.

Access profile: Authenticated client profile.

Parameters

- **proposalId** (*string*) — Proposal ID
- **voter** (*string*) — Voter address
- **vote** (*string*) — Vote type: "for", "against", "abstain"
- **signature** (*string*) — Signed vote

Returns

```
json
{
  "success": true,
  "message": "Vote recorded"
}
```

Security and execution rules

- State-mutating method. Successful execution is bound to the standard authorization envelope and consensus nonce tracking.
- The caller SHOULD obtain a preflight result from `synergy_simulateTransaction` or an equivalent domain-specific simulation step before submission.
- Off-chain signatures do not authorize state mutation on the Synergy Network RPC surface.
- Vote eligibility and counting are determined by the governance snapshot and holding-period rules, not by intra-block token movement.

synergy_executeProposal

```
synergy_executeProposal(proposalId, executor)
```

Execute a passed governance proposal.

Access profile: Authenticated client profile.

Parameters

- **proposalId** (*string*) — Proposal ID
- **executor** (*string*) — Executor address

Returns

json

```
{
  "success": true,
  "message": "Proposal executed",
  "results": [...]
}
```

Security and execution rules

- State-mutating method. Successful execution is bound to the standard authorization envelope and consensus nonce tracking.
- The caller SHOULD obtain a preflight result from `synergy_simulateTransaction` or an equivalent domain-specific simulation step before submission.
- Off-chain signatures do not authorize state mutation on the Synergy Network RPC surface.

synergy_getVotingPower

```
synergy_getVotingPower(address, blockTag)
```

Get the voting power of an address.

Access profile: Public read profile.

Parameters

- **address** (*string*) — Address to query
- **blockTag** (*string, optional*) — Block tag

Returns

json

```
{
  "votingPower": 1000000,
  "delegatedVotingPower": 500000,
  "totalVotingPower": 1500000
}
```

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_delegateVotingPower

```
synergy_delegateVotingPower(delegator, delegate, amount)
```

Delegates governance voting power subject to the protocol holding-period and snapshot rules.

Access profile: Authenticated client profile.

Parameters

- **delegator** (*string*) — Delegator address
- **delegate** (*string*) — Delegate address
- **amount** (*number*) — Amount to delegate (or max if not specified)

Returns

```
json
{
  "success": true,
  "message": "Voting power delegated"
}
```

Security and execution rules

- State-mutating method. Successful execution is bound to the standard authorization envelope and consensus nonce tracking.
- The caller SHOULD obtain a preflight result from `synergy_simulateTransaction` or an equivalent domain-specific simulation step before submission.
- Off-chain signatures do not authorize state mutation on the Synergy Network RPC surface.
- Voting power semantics remain subject to governance snapshots and minimum holding-period rules.

synergy_getTreasuryBalance

```
synergy_getTreasuryBalance()
```

Get the DAO treasury balance.

Access profile: Public read profile.

Parameters

None.

Returns

Treasury object with balances for all tokens

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_proposeTreasurySpend

```
synergy_proposeTreasurySpend(recipient, amount, token, purpose, ...)
```

Propose a treasury expenditure.

Access profile: Authenticated client profile.

Parameters

- **recipient** (*string*) — Recipient address
- **amount** (*number*) — Amount to spend
- **token** (*string*) — Token symbol
- **purpose** (*string*) — Purpose description
- **proposer** (*string*) — Proposer address

Returns

```

json
{
  "success": true,
  "proposalId": "...
}

```

Security and execution rules

- State-mutating method. Successful execution is bound to the standard authorization envelope and consensus nonce tracking.
- The caller SHOULD obtain a preflight result from `synergy_simulateTransaction` or an equivalent domain-specific simulation step before submission.
- Off-chain signatures do not authorize state mutation on the Synergy Network RPC surface.

SXCP, Bridging and Oracles

25 methods

Cross-chain quorum, relay, attestation, bridge, and oracle interfaces.

Methods: `synergy_registerRelayer`, `synergy_unregisterRelayer`, `synergy_relayerHeartbeat`, `synergy_getRelayerSet`, `synergy_getRelayerHealth`, `synergy_getSxcpStatus`, `synergy_submitAttestation`, `synergy_getEventAttestation`, `synergy_getAttestations`, `synergy_slashRelayer`, `synergy_setSxcpHeartbeatTimeout`, `synergy_resetSxcpState`, `synergy_initiateBridgeTransfer`, `synergy_getBridgeTransferStatus`, `synergy_getBridgeConfig`, `synergy_getBridgeLimits`, `synergy_getBridgeFeeEstimate`, `synergy_claimBridgedTokens`, `synergy_getBridgedTokenBalance`, `synergy_getBridgeHistory`, `synergy_submitOracleData`, `synergy_getOracleData`, `synergy_getOracleProviders`, `synergy_registerOracleProvider`, `synergy_getPriceFeed`

`synergy_registerRelayer`

```
synergy_registerRelayer(address, public_key)
```

Register a new SXCP relayer.

Access profile: Authenticated client profile.

Parameters

- **address** (*string*) — Relayer address
- **public_key** (*string*) — Relayer public key (base64 encoded)

Returns

```

json
{
  "success": true,
  "message": "Relayer registered",
  "quorum": { "n": 3, "t": 2 }
}

```

Security and execution rules

- State-mutating method. Successful execution is bound to the standard authorization envelope and consensus nonce tracking.
- The caller SHOULD obtain a preflight result from `synergy_simulateTransaction` or an equivalent domain-specific simulation step before submission.
- Off-chain signatures do not authorize state mutation on the Synergy Network RPC surface.
- Relayer registration participates in the quorum and heartbeat regime used by SXCP verification.

synergy_unregisterRelayer

```
synergy_unregisterRelayer(address)
```

Deactivate a relayer.

Access profile: Authenticated client profile.

Parameters

- **address** (*string*) — Relayer address

Returns

```
json
{
  "success": true,
  "message": "Relayer deactivated",
  "quorum": { "n": 2, "t": 2 }
}
```

Security and execution rules

- State-mutating method. Successful execution is bound to the standard authorization envelope and consensus nonce tracking.
- The caller SHOULD obtain a preflight result from `synergy_simulateTransaction` or an equivalent domain-specific simulation step before submission.
- Off-chain signatures do not authorize state mutation on the Synergy Network RPC surface.

synergy_relayerHeartbeat

```
synergy_relayerHeartbeat(address)
```

Submit a relayer heartbeat.

Access profile: Authenticated client profile.

Parameters

- **address** (*string*) — Relayer address

Returns

```
json
{
  "success": true,
  "message": "Heartbeat recorded",
  "quorum": { "n": 3, "t": 2 }
}
```

Security and execution rules

- State-mutating method. Successful execution is bound to the standard authorization envelope and consensus nonce tracking.
- The caller SHOULD obtain a preflight result from `synergy_simulateTransaction` or an equivalent domain-specific simulation step before submission.
- Off-chain signatures do not authorize state mutation on the Synergy Network RPC surface.

synergy_getRelayerSet

```
synergy_getRelayerSet()
```

Get the current set of registered relayers.

Access profile: Public read profile.

Parameters

None.

Returns

```
json
{
  "relayers": [...],
  "heartbeat_timeout_secs": 120,
  "quorum": { "n": 3, "t": 2 }
}
```

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getRelayerHealth

```
synergy_getRelayerHealth()
```

Get health status of all relayers.

Access profile: Public read profile.

Parameters

None.

Returns

```
json
{
  "relayers": [
    {
      "address": "...",
      "active": true,
      "slashed": false,
      "online": true,
      "heartbeat_age_secs": 30,
      "heartbeat_timeout_secs": 120,
      "eligible_for_quorum": true,
      "reputation": 10,
      "attestation_count": 5
    }
  ],
  "quorum": { "n": 3, "t": 2 }
}
```

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getSxcpStatus

```
synergy_getSxcpStatus()
```

Get overall SXCP status.

Access profile: Public read profile.

Parameters

None.

Returns

```
json
{
  "quorum": { "n": 3, "t": 2 },
  "heartbeat_timeout_secs": 120,
  "relayer_totals": {
    "registered": 5,
    "active": 4,
    "online": 3,
    "slashed": 1
  },
  "event_totals": {
    "tracked": 10,
    "pending": 2,
    "finalized": 8
  },
  "attestation_count": 8,
  "slashing_event_count": 1
}
```

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_submitAttestation

```
synergy_submitAttestation(submitted_by, event_hash, aggregate_sig, metadata)
```

Submit an attestation for a cross-chain event.

Access profile: Authenticated client profile.

Parameters

- **submitted_by** (*string*) — Relayer address submitting the attestation
- **event_hash** (*string*) — Hash of the event being attested
- **aggregate_sig** (*string*) — Aggregate signature (base64 encoded)
- **metadata** (*object, optional*) — Additional metadata

Returns

```
json
{
  "success": true,
  "message": "Attestation support recorded",
  "finalized": false,
  "event_hash": "...",
  "support_count": 1,
}
```



```
"threshold": 2,
"quorum": { "n": 3, "t": 2 },
"timestamp": 1640995200
}
```

Security and execution rules

- State-mutating method. Successful execution is bound to the standard authorization envelope and consensus nonce tracking.
- The caller SHOULD obtain a preflight result from `synergy_simulateTransaction` or an equivalent domain-specific simulation step before submission.
- Off-chain signatures do not authorize state mutation on the Synergy Network RPC surface.
- Attestations are accepted only from eligible relayers and are tallied against the current quorum.

`synergy_getEventAttestation`

```
synergy_getEventAttestation(event_hash)
```

Get attestation status for a specific event.

Access profile: Public read profile.

Parameters

- `event_hash` (*string*) — Event hash

Returns

```
json
{
  "success": true,
  "event": {...},
  "all_supporters": [...],
  "eligible_supporters": [...],
  "support_count": 2,
  "threshold": 2,
  "quorum": { "n": 3, "t": 2 }
}
```

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

`synergy_getAttestations`

```
synergy_getAttestations(limit)
```

Get recent attestations.

Access profile: Public read profile.

Parameters

- `limit` (*number, optional*) — Maximum number of attestations to return (default: 100)

Returns

```
json
{
  "attestations": [...],
}
```

```
"count": 100
}
```

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_slashRelayer

```
synergy_slashRelayer(address, reason, penalty)
```

Slash a relayer for misbehavior.

Access profile: Authenticated client profile.

Parameters

- **address** (*string*) — Relayer address
- **reason** (*string*) — Reason for slashing
- **penalty** (*number, optional*) — Penalty amount (default: 25)

Returns

```
json
{
  "success": true,
  "message": "Relayer slashed",
  "relayer": "...",
  "reason": "...",
  "penalty": 25,
  "newly_finalized_events": [...],
  "quorum": { "n": 2, "t": 2 }
}
```

Security and execution rules

- State-mutating method. Successful execution is bound to the standard authorization envelope and consensus nonce tracking.
- The caller SHOULD obtain a preflight result from synergy_simulateTransaction or an equivalent domain-specific simulation step before submission.
- Off-chain signatures do not authorize state mutation on the Synergy Network RPC surface.

synergy_setSxcpHeartbeatTimeout

```
synergy_setSxcpHeartbeatTimeout(timeout_secs)
```

Set the SXCP heartbeat timeout.

Access profile: Authenticated client profile.

Parameters

- **timeout_secs** (*number*) — Timeout in seconds (minimum: 10)

Returns

```
json
{
  "success": true,
  "heartbeat_timeout_secs": 120,
}
```

```
"quorum": { "n": 3, "t": 2 }
}
```

Security and execution rules

- State-mutating method. Successful execution is bound to the standard authorization envelope and consensus nonce tracking.
- The caller SHOULD obtain a preflight result from `synergy_simulateTransaction` or an equivalent domain-specific simulation step before submission.
- Off-chain signatures do not authorize state mutation on the Synergy Network RPC surface.
- Changing heartbeat policy is an administrative action and SHOULD be tightly role-gated.

`synergy_resetSxcpState`

```
synergy_resetSxcpState(confirmation_token)
```

Reset SXCP state (requires confirmation token).

Access profile: Operator profile.

Parameters

- **confirmation_token** (*string*) — Must be "DEVNET_RESET_SXCP_STATE"

Returns

```
json
{
  "success": true,
  "message": "SXCP state reset"
}
```

Security and execution rules

- Operator-profile method. Expose only on authenticated administrative control planes.
- If a deployment offers this method remotely, it SHOULD be protected by mutually authenticated transport, role-based policy, and audit logging.
- This is an administrative reset path and is unsuitable for public exposure.

`synergy_initiateBridgeTransfer`

```
synergy_initiateBridgeTransfer(fromAddress, toAddress, token, amount, ...)
```

Initiates a cross-chain transfer under SXCP verification and protocol delay rules.

Access profile: Authenticated client profile.

Parameters

- **fromAddress** (*string*) — Source address
- **toAddress** (*string*) — Destination address on target chain
- **token** (*string*) — Token to bridge
- **amount** (*number*) — Amount to bridge
- **targetChain** (*string*) — Target chain identifier
- **initiator** (*string*) — Initiator address

Returns

```
json
{
  "success": true,
```

```

    "bridgeTransferId": "...",
    "estimatedCompletionTime": 1640995200,
    "fee": 1000000
  }

```

Security and execution rules

- State-mutating method. Successful execution is bound to the standard authorization envelope and consensus nonce tracking.
- The caller SHOULD obtain a preflight result from `synergy_simulateTransaction` or an equivalent domain-specific simulation step before submission.
- Off-chain signatures do not authorize state mutation on the Synergy Network RPC surface.
- Large transfers may enter protocol delay windows and require native attestation verification before release.

`synergy_getBridgeTransferStatus`

```
synergy_getBridgeTransferStatus(bridgeTransferId)
```

Get status of a bridge transfer.

Access profile: Public read profile.

Parameters

- `bridgeTransferId` (*string*) — Bridge transfer ID

Returns

Status object with current state and progress

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

`synergy_getBridgeConfig`

```
synergy_getBridgeConfig(targetChain)
```

Get bridge configuration for a target chain.

Access profile: Public read profile.

Parameters

- `targetChain` (*string*) — Target chain identifier

Returns

Bridge configuration object

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

`synergy_getBridgeLimits`

```
synergy_getBridgeLimits(targetChain, token)
```

Get bridge transfer limits.

Access profile: Public read profile.

Parameters

- **targetChain** (*string*)— Target chain identifier
- **token** (*string*)— Token symbol

Returns

Limits object with min/max amounts and daily limits

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getBridgeFeeEstimate

```
synergy_getBridgeFeeEstimate(targetChain, token, amount)
```

Estimate bridge transfer fees.

Access profile: Public read profile.

Parameters

- **targetChain** (*string*)— Target chain identifier
- **token** (*string*)— Token symbol
- **amount** (*number*)— Amount to bridge

Returns

Fee estimate object

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_claimBridgedTokens

```
synergy_claimBridgedTokens(bridgeTransferId, claimer, proof)
```

Claims a finalized cross-chain transfer using the required attestation proof set.

Access profile: Authenticated client profile.

Parameters

- **bridgeTransferId** (*string*)— Bridge transfer ID
- **claimer** (*string*)— Claimer address
- **proof** (*string*)— Attestation proof

Returns

```
json
{
  "success": true,
  "transactionHash": "..."
```

Security and execution rules

- State-mutating method. Successful execution is bound to the standard authorization envelope and consensus nonce tracking.
- The caller SHOULD obtain a preflight result from synergy_simulateTransaction or an equivalent domain-specific simulation step before submission.
- Off-chain signatures do not authorize state mutation on the Synergy Network RPC surface.
- Claims are valid only when the attestation set satisfies the relevant SXCP threshold and finality policy.

synergy_getBridgedTokenBalance

```
synergy_getBridgedTokenBalance(address, originalChain, token)
```

Get balance of bridged tokens.

Access profile: Public read profile.

Parameters

- **address** (*string*) — Address to query
- **originalChain** (*string*) — Original chain
- **token** (*string*) — Token symbol

Returns

Bridged token balance

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getBridgeHistory

```
synergy_getBridgeHistory(address, direction, limit)
```

Get bridge transfer history.

Access profile: Public read profile.

Parameters

- **address** (*string*) — Address to query
- **direction** (*string, optional*) — "inbound" or "outbound"
- **limit** (*number, optional*) — Maximum results

Returns

Array of bridge transfer objects

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_submitOracleData

```
synergy_submitOracleData(oracleAddress, dataId, value, timestamp, ...)
```

Submits oracle data signed by an authorized oracle provider.

Access profile: Authenticated client profile.

Parameters

- **oracleAddress** (*string*) — Oracle address
- **dataId** (*string*) — Data identifier
- **value** (*any*) — Data value
- **timestamp** (*number*) — Data timestamp
- **signature** (*string*) — Oracle signature

Returns

```
json
{
```

```

    "success": true,
    "message": "Oracle data submitted"
  }

```

Security and execution rules

- State-mutating method. Successful execution is bound to the standard authorization envelope and consensus nonce tracking.
- The caller SHOULD obtain a preflight result from `synergy_simulateTransaction` or an equivalent domain-specific simulation step before submission.
- Off-chain signatures do not authorize state mutation on the Synergy Network RPC surface.
- Oracle submissions are accepted only from registered providers with valid signatures.

`synergy_getOracleData`

```
synergy_getOracleData(dataId)
```

Get the latest oracle data.

Access profile: Public read profile.

Parameters

- `dataId` (*string*) — Data identifier

Returns

Oracle data object with value and metadata

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

`synergy_getOracleProviders`

```
synergy_getOracleProviders()
```

Get all registered oracle providers.

Access profile: Public read profile.

Parameters

None.

Returns

Array of oracle provider objects

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

`synergy_registerOracleProvider`

```
synergy_registerOracleProvider(address, metadata, deposit)
```

Register a new oracle provider.

Access profile: Authenticated client profile.

Parameters

- `address` (*string*) — Oracle address

- **metadata** (*object*)— Oracle metadata
- **deposit** (*number*)— Security deposit

Returns

```
json
{
  "success": true,
  "message": "Oracle provider registered"
}
```

Security and execution rules

- State-mutating method. Successful execution is bound to the standard authorization envelope and consensus nonce tracking.
- The caller SHOULD obtain a preflight result from `synergy_simulateTransaction` or an equivalent domain-specific simulation step before submission.
- Off-chain signatures do not authorize state mutation on the Synergy Network RPC surface.

`synergy_getPriceFeed`

```
synergy_getPriceFeed(pair)
```

Get price feed data.

Access profile: Public read profile.

Parameters

- **pair** (*string*)— Trading pair (e.g., "SNRG/USD")

Returns

Price feed object with current price and history

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

Analytics, Monitoring and Security Telemetry

20 methods

Historical analytics, gas tracking, node health, metrics, logs, and security telemetry surfaces.

Methods: `synergy_getAddressTransactions`, `synergy_getAddressInternalTransactions`, `synergy_getAddressBalanceHistory`, `synergy_getTokenHolders`, `synergy_getTokenTransfers`, `synergy_getNetworkActivity`, `synergy_getTransactionVolume`, `synergy_getActiveAddresses`, `synergy_getContractInteractions`, `synergy_getGasTracker`, `synergy_getHealthCheck`, `synergy_getMetrics`, `synergy_getDebugInfo`, `synergy_getConfig`, `synergy_reloadConfig`, `synergy_getLogLevels`, `synergy_setLogLevel`, `synergy_getSecurityAudit`, `synergy_getPrivateTransactionStatus`, `synergy_getPrivacyPoolStats`

`synergy_getAddressTransactions`

```
synergy_getAddressTransactions(address, fromBlock, toBlock, limit)
```

Get all transactions for an address.

Access profile: Public read profile.

Parameters

- **address** (*string*)— Address to query

- **fromBlock** (*number, optional*) — Starting block
- **toBlock** (*number, optional*) — Ending block
- **limit** (*number, optional*) — Maximum results

Returns

Array of transaction objects

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getAddressInternalTransactions

```
synergy_getAddressInternalTransactions(address, fromBlock, toBlock)
```

Get internal transactions for an address.

Access profile: Public read profile.

Parameters

- **address** (*string*) — Address to query
- **fromBlock** (*number, optional*) — Starting block
- **toBlock** (*number, optional*) — Ending block

Returns

Array of internal transaction objects

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getAddressBalanceHistory

```
synergy_getAddressBalanceHistory(address, fromBlock, toBlock)
```

Get balance history for an address.

Access profile: Public read profile.

Parameters

- **address** (*string*) — Address to query
- **fromBlock** (*number, optional*) — Starting block
- **toBlock** (*number, optional*) — Ending block

Returns

Array of balance snapshots

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getTokenHolders

```
synergy_getTokenHolders(tokenSymbol, limit, offset)
```

Get all holders of a token.

Access profile: Public read profile.

Parameters

- **tokenSymbol** (*string*)— Token symbol
- **limit** (*number, optional*)— Maximum results
- **offset** (*number, optional*)— Offset for pagination

Returns

- Array of holder objects with:
- address: Holder address
- balance: Token balance
- percentage: Percentage of total supply

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getTokenTransfers

```
synergy_getTokenTransfers(tokenSymbol, fromBlock, toBlock, limit)
```

Get all transfers of a specific token.

Access profile: Public read profile.

Parameters

- **tokenSymbol** (*string*)— Token symbol
- **fromBlock** (*number, optional*)— Starting block
- **toBlock** (*number, optional*)— Ending block
- **limit** (*number, optional*)— Maximum results

Returns

Array of transfer objects

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getNetworkActivity

```
synergy_getNetworkActivity(period)
```

Get network activity statistics.

Access profile: Public read profile.

Parameters

- **period** (*string, optional*)— Time period: "1h", "24h", "7d", "30d"

Returns

- Activity object with:
- totalTransactions: Total transactions
- uniqueAddresses: Unique active addresses
- averageBlockTime: Average block time
- averageGasPrice: Average gas price
- totalVolume: Total transaction volume

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getTransactionVolume

```
synergy_getTransactionVolume(period, token)
```

Get transaction volume statistics.

Access profile: Public read profile.

Parameters

- **period** (*string, optional*) — Time period
- **token** (*string, optional*) — Specific token

Returns

Volume statistics with hourly/daily breakdown

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getActiveAddresses

```
synergy_getActiveAddresses(period)
```

Get count of active addresses.

Access profile: Public read profile.

Parameters

- **period** (*string, optional*) — Time period

Returns

- Object with:
- **dailyActive**: Daily active addresses
- **weeklyActive**: Weekly active addresses
- **monthlyActive**: Monthly active addresses

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getContractInteractions

```
synergy_getContractInteractions(contractAddress, period)
```

Get contract interaction statistics.

Access profile: Public read profile.

Parameters

- **contractAddress** (*string*) — Contract address
- **period** (*string, optional*) — Time period

Returns

Interaction statistics

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getGasTracker

```
synergy_getGasTracker()
```

Get real-time gas tracking information.

Access profile: Public read profile.

Parameters

None.

Returns

- Gas tracker object with:
- slow: Slow gas price and estimated time
- average: Average gas price and estimated time
- fast: Fast gas price and estimated time
- baseFee: Current base fee

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getHealthCheck

```
synergy_getHealthCheck()
```

Get comprehensive node health check.

Access profile: Public read profile.

Parameters

None.

Returns

- Health object with:
- status: Overall status ("healthy", "degraded", "unhealthy")
- blockchain: Blockchain sync status
- p2p: P2P network status
- rpc: RPC server status
- memory: Memory usage
- disk: Disk usage
- cpu: CPU usage

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getMetrics

```
synergy_getMetrics()
```

Get Prometheus-compatible metrics.

Access profile: Operator profile.

Parameters

None.

Returns

Metrics in Prometheus format

Security and execution rules

- Operator-profile method. Expose only on authenticated administrative control planes.
- If a deployment offers this method remotely, it SHOULD be protected by mutually authenticated transport, role-based policy, and audit logging.

synergy_getDebugInfo

```
synergy_getDebugInfo(category)
```

Get debug information for troubleshooting.

Access profile: Operator profile.

Parameters

- **category** (*string, optional*) — Debug category

Returns

Debug information object

Security and execution rules

- Operator-profile method. Expose only on authenticated administrative control planes.
- If a deployment offers this method remotely, it SHOULD be protected by mutually authenticated transport, role-based policy, and audit logging.

synergy_getConfig

```
synergy_getConfig()
```

Get the current node configuration.

Access profile: Operator profile.

Parameters

None.

Returns

Configuration object (sensitive data redacted)

Security and execution rules

- Operator-profile method. Expose only on authenticated administrative control planes.
- If a deployment offers this method remotely, it SHOULD be protected by mutually authenticated transport, role-based policy, and audit logging.

synergy_reloadConfig

```
synergy_reloadConfig()
```

Reload node configuration.

Access profile: Operator profile.

Parameters

None.

Returns

```
json
{
  "success": true,
  "message": "Configuration reloaded"
```

```
}
```

Security and execution rules

- Operator-profile method. Expose only on authenticated administrative control planes.
- If a deployment offers this method remotely, it SHOULD be protected by mutually authenticated transport, role-based policy, and audit logging.

synergy_getLogLevels

```
synergy_getLogLevels()
```

Get current log levels.

Access profile: Operator profile.

Parameters

None.

Returns

Object with current log levels for each module

Security and execution rules

- Operator-profile method. Expose only on authenticated administrative control planes.
- If a deployment offers this method remotely, it SHOULD be protected by mutually authenticated transport, role-based policy, and audit logging.

synergy_setLogLevel

```
synergy_setLogLevel(module, level)
```

Set log level for a module.

Access profile: Operator profile.

Parameters

- **module** (*string*)— Module name
- **level** (*string*)— Log level: "error", "warn", "info", "debug", "trace"

Returns

```
json
{
  "success": true,
  "message": "Log level updated"
}
```

Security and execution rules

- Operator-profile method. Expose only on authenticated administrative control planes.
- If a deployment offers this method remotely, it SHOULD be protected by mutually authenticated transport, role-based policy, and audit logging.

synergy_getSecurityAudit

```
synergy_getSecurityAudit()
```

Get security audit information for the node.

Access profile: Operator profile.

Parameters

None.

Returns

- Security audit object with:
- lastAudit: Last audit timestamp
- vulnerabilities: Known vulnerabilities
- recommendations: Security recommendations
- complianceStatus: Compliance status

Security and execution rules

- Operator-profile method. Expose only on authenticated administrative control planes.
- If a deployment offers this method remotely, it SHOULD be protected by mutually authenticated transport, role-based policy, and audit logging.

synergy_getPrivateTransactionStatus

```
synergy_getPrivateTransactionStatus(txHash)
```

Get status of a private transaction.

Access profile: Public read profile.

Parameters

- txHash (*string*) — Transaction hash

Returns

Private transaction status

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getPrivacyPoolStats

```
synergy_getPrivacyPoolStats()
```

Get statistics about the privacy pool.

Access profile: Public read profile.

Parameters

None.

Returns

Pool statistics

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

Developer, Batch and Operator Controls

14 methods

Batch execution and isolated execution profile methods for testing, protocol validation, and operator workflows.

Methods: `synergy_mine`, `synergy_setAccountBalance`, `synergy_impersonateAccount`, `synergy_stopImpersonatingAccount`, `synergy_snapshot`, `synergy_revertToSnapshot`, `synergy_reset`, `synergy_setStorageAt`, `synergy_setCode`, `synergy_setNonce`, `synergy_batch`, `synergy_multicall`, `synergy_aggregate`, `synergy_emergencyStop`

`synergy_mine`

```
synergy_mine(count)
```

Produces one or more blocks in an isolated execution profile used for controlled testing and protocol validation.

Access profile: Operator profile.

Parameters

- **count** (*number, optional*) — Number of blocks to mine

Returns

Array of mined block hashes

Security and execution rules

- Operator-profile method. Expose only on authenticated administrative control planes.
- If a deployment offers this method remotely, it SHOULD be protected by mutually authenticated transport, role-based policy, and audit logging.
- This method belongs on isolated execution profiles only.

`synergy_setAccountBalance`

```
synergy_setAccountBalance(address, balance)
```

Sets an account balance inside an isolated execution profile for controlled testing and protocol validation.

Access profile: Operator profile.

Parameters

- **address** (*string*) — Address
- **balance** (*number*) — New balance

Returns

```
json
{
  "success": true,
  "message": "Balance updated"
}
```

Security and execution rules

- Operator-profile method. Expose only on authenticated administrative control planes.
- If a deployment offers this method remotely, it SHOULD be protected by mutually authenticated transport, role-based policy, and audit logging.
- This method belongs on isolated execution profiles only.

`synergy_impersonateAccount`

```
synergy_impersonateAccount(address)
```

Activates account impersonation inside an isolated execution profile for controlled testing and protocol validation.

Access profile: Operator profile.

Parameters

- **address** (*string*) — Address to impersonate

Returns

```
json
{
  "success": true,
  "message": "Account impersonated"
}
```

Security and execution rules

- Operator-profile method. Expose only on authenticated administrative control planes.
- If a deployment offers this method remotely, it SHOULD be protected by mutually authenticated transport, role-based policy, and audit logging.
- This method belongs on isolated execution profiles only.

synergy_stopImpersonatingAccount

```
synergy_stopImpersonatingAccount(address)
```

Stops an impersonation session inside an isolated execution profile.

Access profile: Operator profile.

Parameters

- **address** (*string*) — Address to stop impersonating

Returns

```
json
{
  "success": true,
  "message": "Stopped impersonating"
}
```

Security and execution rules

- Operator-profile method. Expose only on authenticated administrative control planes.
- If a deployment offers this method remotely, it SHOULD be protected by mutually authenticated transport, role-based policy, and audit logging.
- This method belongs on isolated execution profiles only.

synergy_snapshot

```
synergy_snapshot()
```

Takes an isolated execution profile state snapshot for repeatable testing workflows.

Access profile: Operator profile.

Parameters

None.

Returns

```
json
{
  "success": true,
  "snapshotId": "..."
```

```
}
```

Security and execution rules

- Operator-profile method. Expose only on authenticated administrative control planes.
- If a deployment offers this method remotely, it SHOULD be protected by mutually authenticated transport, role-based policy, and audit logging.
- This method belongs on isolated execution profiles only.

synergy_revertToSnapshot

```
synergy_revertToSnapshot(snapshotId)
```

Reverts an isolated execution profile to a previously captured snapshot.

Access profile: Operator profile.

Parameters

- **snapshotId** (*string*)— Snapshot ID

Returns

```
json
{
  "success": true,
  "message": "State reverted"
}
```

Security and execution rules

- Operator-profile method. Expose only on authenticated administrative control planes.
- If a deployment offers this method remotely, it SHOULD be protected by mutually authenticated transport, role-based policy, and audit logging.
- This method belongs on isolated execution profiles only.

synergy_reset

```
synergy_reset(blockNumber, jsonConfig)
```

Resets an isolated execution profile to a specific block or supplied genesis configuration.

Access profile: Operator profile.

Parameters

- **blockNumber** (*number*)— Block number to reset to, OR
- **jsonConfig** (*object*)— Genesis configuration

Returns

```
json
{
  "success": true,
  "message": "Blockchain reset"
}
```

Security and execution rules

- Operator-profile method. Expose only on authenticated administrative control planes.
- If a deployment offers this method remotely, it SHOULD be protected by mutually authenticated transport, role-based policy, and audit logging.

- This method belongs on isolated execution profiles only.

synergy_setStorageAt

```
synergy_setStorageAt(address, slot, value)
```

Mutates storage at a specific slot inside an isolated execution profile.

Access profile: Operator profile.

Parameters

- **address** (*string*) — Contract address
- **slot** (*string*) — Storage slot (hex)
- **value** (*string*) — Value to set (hex)

Returns

```
json
{
  "success": true,
  "message": "Storage updated"
}
```

Security and execution rules

- Operator-profile method. Expose only on authenticated administrative control planes.
- If a deployment offers this method remotely, it SHOULD be protected by mutually authenticated transport, role-based policy, and audit logging.
- This method belongs on isolated execution profiles only.

synergy_setCode

```
synergy_setCode(address, code)
```

Writes code at an address inside an isolated execution profile.

Access profile: Operator profile.

Parameters

- **address** (*string*) — Address
- **code** (*string*) — Contract bytecode (hex)

Returns

```
json
{
  "success": true,
  "message": "Code updated"
}
```

Security and execution rules

- Operator-profile method. Expose only on authenticated administrative control planes.
- If a deployment offers this method remotely, it SHOULD be protected by mutually authenticated transport, role-based policy, and audit logging.
- This method belongs on isolated execution profiles only.

synergy_setNonce

```
synergy_setNonce(address, nonce)
```

Sets the nonce for an address inside an isolated execution profile.

Access profile: Operator profile.

Parameters

- **address** (*string*) — Address
- **nonce** (*number*) — New nonce

Returns

```
json
{
  "success": true,
  "message": "Nonce updated"
}
```

Security and execution rules

- Operator-profile method. Expose only on authenticated administrative control planes.
- If a deployment offers this method remotely, it SHOULD be protected by mutually authenticated transport, role-based policy, and audit logging.
- This method belongs on isolated execution profiles only.

synergy_batch

```
synergy_batch(calls, authorization)
```

Submits a batch of RPC calls under a single request envelope while preserving per-call authorization and manifest review.

Access profile: Authenticated client profile.

Parameters

- **calls** (*array*) — Array of RPC call objects.
- **authorization** (*array, optional*) — Per-call AuthorizationSignatures for mutating calls.

Returns

Array of per-call results in request order.

Security and execution rules

- State-mutating method. Successful execution is bound to the standard authorization envelope and consensus nonce tracking.
- The caller SHOULD obtain a preflight result from `synergy_simulateTransaction` or an equivalent domain-specific simulation step before submission.
- Off-chain signatures do not authorize state mutation on the Synergy Network RPC surface.
- Each mutating sub-call retains its own authorization requirements and manifest review.

synergy_multicall

```
synergy_multicall(calls, blockTag)
```

Executes multiple contract calls in one envelope with per-sub-call manifest review and linear-value enforcement.

Access profile: Authenticated client profile.

Parameters

- **calls** (*array*)— Array of contract call objects.
- **blockTag** (*string, optional*)— State reference for read aggregation or simulation.

Returns

Array of call results in request order.

Security and execution rules

- State-mutating method. Successful execution is bound to the standard authorization envelope and consensus nonce tracking.
- The caller SHOULD obtain a preflight result from `synergy_simulateTransaction` or an equivalent domain-specific simulation step before submission.
- Off-chain signatures do not authorize state mutation on the Synergy Network RPC surface.
- Per-sub-call manifests and linear value semantics prevent hidden value reuse across sub-calls.
- Deployments SHOULD enforce a bounded sub-call count to preserve readability and safety.

synergy_aggregate

```
synergy_aggregate(queries, aggregation)
```

Aggregates multiple RPC queries using a declared aggregation function and returns a single composite result.

Access profile: Authenticated client profile.

Parameters

- **queries** (*array*)— Array of queries
- **aggregation** (*string*)— Aggregation function

Returns

Aggregated result

Security and execution rules

- State-mutating method. Successful execution is bound to the standard authorization envelope and consensus nonce tracking.
- The caller SHOULD obtain a preflight result from `synergy_simulateTransaction` or an equivalent domain-specific simulation step before submission.
- Off-chain signatures do not authorize state mutation on the Synergy Network RPC surface.

synergy_emergencyStop

```
synergy_emergencyStop(reason, authorizedBy)
```

Invokes the emergency stop control path for authorized network stewards.

Access profile: Operator profile.

Parameters

- **reason** (*string*)— Reason for emergency stop
- **authorizedBy** (*string*)— Authorizing address

Returns

json

```
{
  "success": true,
  "message": "Emergency stop initiated"
```

}

Security and execution rules

- Operator-profile method. Expose only on authenticated administrative control planes.
- If a deployment offers this method remotely, it SHOULD be protected by mutually authenticated transport, role-based policy, and audit logging.
- Emergency controls SHOULD be protected by the strongest available operator authorization posture and auditable quorum policy.

Protocol Utilities, Supply and Notifications

14 methods

Epoch, supply, constants, schedule, notification, and compatibility surfaces that round out the canonical RPC manual.

Methods: `synergy_getEpochInfo`, `synergy_getHistoricalEpoch`, `synergy_getInflationRate`, `synergy_getCirculatingSupply`, `synergy_getTokenSupply`, `synergy_getProtocolVersion`, `synergy_getForkSchedule`, `synergy_getSpec`, `synergy_getConstants`, `synergy_getUpcomingEvents`, `synergy_registerWebhook`, `synergy_unregisterWebhook`, `synergy_getUncleByBlockNumberAndIndex`, `synergy_getUncleCountByBlockNumber`

`synergy_getEpochInfo``synergy_getEpochInfo()`

Get current epoch information.

Access profile: Public read profile.

Parameters

None.

Returns

- Epoch object with:
- `epoch`: Current epoch number
- `startTime`: Epoch start timestamp
- `endTime`: Epoch end timestamp
- `totalValidators`: Validators in epoch
- `totalBlocks`: Blocks in epoch
- `totalRewards`: Total rewards distributed

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

`synergy_getHistoricalEpoch``synergy_getHistoricalEpoch(epoch)`

Get information about a historical epoch.

Access profile: Public read profile.

Parameters

- `epoch` (*number*) — Epoch number

Returns

Historical epoch information

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getInflationRate

```
synergy_getInflationRate()
```

Get current inflation rate.

Access profile: Public read profile.

Parameters

None.

Returns

```
json
{
  "currentInflationRate": 0.05,
  "annualInflationRate": 0.05,
  "nextAdjustmentAt": 1640995200
}
```

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getCirculatingSupply

```
synergy_getCirculatingSupply()
```

Get circulating token supply.

Access profile: Public read profile.

Parameters

None.

Returns

```
json
{
  "circulatingSupply": 1000000000,
  "totalSupply": 1200000000,
  "stakedSupply": 500000000,
  "lockedSupply": 100000000
}
```

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getTokenSupply

```
synergy_getTokenSupply(token)
```

Get token supply information.

Access profile: Public read profile.

Parameters

- **token** (*string, optional*)— Token symbol (default: "SNRG")

Returns

Supply breakdown object

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getProtocolVersion

```
synergy_getProtocolVersion()
```

Get the protocol version.

Access profile: Public read profile.

Parameters

None.

Returns

json

```
{
  "version": "1.0.0",
  "minCompatibleVersion": "0.9.0",
  "recommendedVersion": "1.0.0"
}
```

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getForkSchedule

```
synergy_getForkSchedule()
```

Get upcoming fork schedule.

Access profile: Public read profile.

Parameters

None.

Returns

- Array of scheduled forks with:
 - name: Fork name
 - block: Activation block
 - features: New features
 - status: "scheduled", "active"

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getSpec

```
synergy_getSpec()
```

Get network specification parameters.

Access profile: Public read profile.

Parameters

None.

Returns

Spec object with all network constants

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getConstants

```
synergy_getConstants()
```

Get network constants and parameters.

Access profile: Public read profile.

Parameters

None.

Returns

Object with network constants

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_getUpcomingEvents

```
synergy_getUpcomingEvents(limit)
```

Get upcoming network events.

Access profile: Public read profile.

Parameters

- **limit** (*number, optional*) — Maximum events to return

Returns

Array of event objects

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.

synergy_registerWebhook

```
synergy_registerWebhook(url, events, secret)
```

Register a webhook for event notifications.

Access profile: Authenticated client profile.

Parameters

- **url** (*string*)— Webhook URL
- **events** (*array*)— Events to subscribe to
- **secret** (*string, optional*)— Webhook secret

Returns

```
json
{
  "success": true,
  "webhookId": "..."
```

Security and execution rules

- State-mutating method. Successful execution is bound to the standard authorization envelope and consensus nonce tracking.
- The caller SHOULD obtain a preflight result from `synergy_simulateTransaction` or an equivalent domain-specific simulation step before submission.
- Off-chain signatures do not authorize state mutation on the Synergy Network RPC surface.

`synergy_unregisterWebhook`

```
synergy_unregisterWebhook(webhookId)
```

Unregister a webhook.

Access profile: Authenticated client profile.

Parameters

- **webhookId** (*string*)— Webhook ID

Returns

```
json
{
  "success": true,
  "message": "Webhook unregistered"
```

Security and execution rules

- State-mutating method. Successful execution is bound to the standard authorization envelope and consensus nonce tracking.
- The caller SHOULD obtain a preflight result from `synergy_simulateTransaction` or an equivalent domain-specific simulation step before submission.
- Off-chain signatures do not authorize state mutation on the Synergy Network RPC surface.

`synergy_getUncleByBlockNumberAndIndex`

```
synergy_getUncleByBlockNumberAndIndex(blockNumber, uncleIndex)
```

Returns the uncle-like compatibility view for a block index. On Synergy consensus profiles without uncle semantics, the method returns null.

Access profile: Public read profile.

Parameters

- blockNumber (number)
- uncleIndex (number)

Returns

Uncle block object or null

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.
- This is a compatibility surface. Consensus profiles without uncle semantics return null.

synergy_getUncleCountByBlockNumber

```
synergy_getUncleCountByBlockNumber(blockNumber)
```

Returns the uncle count compatibility value for a block. On Synergy consensus profiles without uncle semantics, the method returns 0.

Access profile: Public read profile.

Parameters

- blockNumber (number)

Returns

number - Uncle count (currently always 0)

Security and execution rules

- Read-only method. It does not change consensus state.
- Deployments may expose this surface on public read profiles unless narrowed by local policy.
- This is a compatibility surface. Consensus profiles without uncle semantics return 0.